



Innovation & entrepreneurship Hub for for educated rural youth(SURE Trust-IERY)



Trust-IERY)

Innovation & entrepreneurship Hub for for educated rural youth(SURE

ASSIGNMENT 3
V.MOUNIKA
G4-25-VLSI
THE DOMAIN OF THE
PROJECT
PHYSICAL DESIGN



Assignment: RISC-V 32-bit RTL Physical Design Flow

1. Objective

To implement the RISC-V 32-bit RTL design using OpenLane flow and apply timing, floorplan, placement, and CTS constraints.

2. Tools Used

- Linux environment / Ubuntu
- Docker
- OpenLane
- OpenROAD
- Yosys (RTL synthesis)
- OpenSTA (Static Timing Analysis)
- Magic (Layout viewer)
- Sky130 PDK
- RISC-V 32-bit RTL source code

3. Required Design Files

Example:

```
designs/  
└─ riscv32/  
    └─ config.tcl  
    └─ src/  
        └─ picorv32.v  
    └─ pin_order.cfg
```

4. Starting OpenLane Environment

Commands:

```
cd OpenLane
```

```
make mount
```

Enter OpenLane container:

```
./flow.tcl -interactive
```

or

```
docker run ...
```

5. Preparing Design Configuration

Edit:



```
designs/riscv32/config.tcl
```

Add clock information:

```
set ::env(CLOCK_PORT) clk
set ::env(CLOCK_PERIOD) 10
```

6. Applying Assignment Constraints

6.1 Clock Uncertainty

For synthesis:

```
set_clock_uncertainty 15% of clock period
```

Example:

Clock period = 10ns

```
set_clock_uncertainty 1.5
```

For CTS:

```
set_clock_uncertainty 0.5
```

(5% of 10ns)

6.2 Clock Derate

Set:

```
set_timing_derate 3%
```

Used to model variation in delay.

6.3 Input and Output Delay

50% of clock period:

For 10ns clock:

Input delay:

```
set_input_delay 5
```

Output delay:

```
set_output_delay 5
```



7. Floorplan Settings

Utilization

Set:

```
FP_CORE_UTIL = 60
```

Meaning:

60% of core area is occupied by standard cells.

Floorplan Ratio

Set:

```
FP_ASPECT_RATIO = 0.7
```

8. Port Placement

Requirement:

- Inputs on one side
- Outputs on opposite side
- Metal4 layer

File:

```
pin_order.cfg
```

Example:

```
N
clk
reset
```

```
S
out_data
```

Layer:

```
FP_PIN_ORDER_CFG
```

and

```
FP_IO_HLAYER = metal4
FP_IO_VLAYER = metal4
```



9. Enable Core Ring

In config.tcl:

```
FP_PDN_CORE_RING = 1
```

Purpose:

- Provides power distribution
- Reduces IR drop

10. CTS Configuration

Target skew:

5% of clock period

Example:

10ns clock:

```
CTS_TARGET_SKEW = 0.5
```

Enable clock network delay:

```
CTS_CLK_MAX_WIRE_LENGTH
```

11. Running OpenLane Flow

Inside OpenLane:

Synthesis

```
./flow.tcl -design riscv32 -tag run1 -to synth
```

Floorplan

```
./flow.tcl -design riscv32 -tag run1 -to floorplan
```

Placement

```
./flow.tcl -design riscv32 -tag run1 -to placement
```

CTS

```
./flow.tcl -design riscv32 -tag run1 -to cts
```



Routing

```
./flow.tcl -design riscv32 -tag run1
```

12. Viewing Layout

Open Magic:

```
magic -T /pdk/sky130A/libs.tech/magic/sky130A.tech \
designs/riscv32/runs/run1/results/final/def/
```

Check:

- Standard cells
- Power rings
- Clock tree
- Routing

13. Results to Capture

Screenshots:

1. Synthesis completed
2. Floorplan view
3. Placement view
4. CTS clock tree
5. Final routed layout
6. Timing report

1. RISC-V 32-bit RTL Program (Verilog)

Usually the RTL file used is:

```
picorv32.v
```

This contains the complete RISC-V 32-bit processor RTL.

It includes:

- Program counter (PC)
 - Instruction decoder
 - ALU
 - Register file
 - Control logic
 - Memory interface
 - Branch/jump logic
- The top module is generally:



```
module picorv32 (  
    input clk,  
    input resetn,  
    output mem_valid,  
    output mem_instr,  
    output [31:0] mem_addr,  
    output [31:0] mem_wdata,  
    input  [31:0] mem_rdata,  
    input mem_ready  
);
```

This file is placed inside:

designs/riscv32/src/

Example structure:

```
riscv32  
├── src  
│   └── picorv32.v  
├── config.tcl  
└── pin_order.cfg
```

2. OpenLane Configuration Program

File:

config.tcl

This controls the physical design constraints.

Used parameters:

```
set ::env(DSIGN_NAME) picorv32  
  
set ::env(VERILOG_FILES) \  
$::env(DSIGN_DIR)/src/picorv32.v  
  
set ::env(CLOCK_PORT) clk  
  
set ::env(CLOCK_PERIOD) 10
```

3. Synthesis Timing Constraints

Inside config.tcl:

Clock uncertainty



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```
set ::env(SYNTH_CLOCK_UNCERTAINTY) 1.5
```

(15% of 10ns clock period)

CTS uncertainty:

```
set ::env(CTS_CLK_UNCERTAINTY) 0.5
```

(5% of 10ns clock period)

4. Timing Derate

```
set ::env(SYNTH_TIMING_DERATE) 0.03
```

(3% derate)

5. Input and Output Delay Constraints

Input delay:

```
set ::env(SDC_INPUT_DELAY) 5
```

Output delay:

```
set ::env(SDC_OUTPUT_DELAY) 5
```

(50% of 10ns clock period)

6. Floorplan Constraints

60% utilization:

```
set ::env(FP_CORE_UTIL) 60
```

Floorplan ratio:

```
set ::env(FP_ASPECT_RATIO) 0.7
```

7. Pin Placement File

File:

```
pin_order.cfg
```

Used for placing inputs and outputs.



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Example format:

```
# Input pins

clk
resetsn

# Output pins

mem_valid
mem_instr
mem_addr
mem_wdata
```

8. Core Ring Enable

In:

```
config.tcl
```

Parameter:

```
set ::env(FP_PDN_CORE_RING) 1
```

9. CTS Configuration

Target skew:

```
set ::env(CTS_TARGET_SKEW) 0.5
```

Clock network insertion:

```
set ::env(CTS_CLK_MAX_WIRE_LENGTH) 100
```

10. OpenLane Execution Commands Used

Mount OpenLane:

```
cd OpenLane
make mount
```

Run complete flow:

```
./flow.tcl -design riscv32 -tag run1
```

Run individual stages:

Synthesis

```
./flow.tcl -design riscv32 -tag run1 -to synth
```



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Floorplan

```
./flow.tcl -design riscv32 -tag run1 -to floorplan
```

Placement

```
./flow.tcl -design riscv32 -tag run1 -to placement
```

CTS

```
./flow.tcl -design riscv32 -tag run1 -to cts
```

Routing

```
./flow.tcl -design riscv32 -tag run1
```

"Programs Used / Files Used" section:

1. picorv32.v (RTL source)
2. config.tcl (OpenLane constraints)
3. pin_order.cfg (port placement)
4. OpenLane TCL flow commands

Program 1: picorv32.v (RISC-V 32-bit RTL Source)

File Name:

picorv32.v

Purpose:

- This is the main RTL (Verilog) source code of the RISC-V 32-bit processor.
- It describes the hardware functionality before synthesis.
- OpenLane takes this file as input and converts it into a gate-level netlist.

Location in OpenLane design:

designs/riscv32/src/picorv32.v

Top Module:

```
module picorv32 (  
    input clk,  
    input resetn,  
  
    output reg mem_valid,  
    output reg mem_instr,  
  
    output reg [31:0] mem_addr,  
    output reg [31:0] mem_wdata,
```



```
    input [31:0] mem_rdata,  
    input mem_ready  
);
```

Main blocks inside `picorv32.v`

1. Register Declaration

```
reg [31:0] pc;  
reg [31:0] next_pc;  
  
reg [31:0] registers [0:31];  
  
reg [31:0] instr;
```

Function:

- `pc` → Stores current program counter address.
- `registers` → 32 general-purpose RISC-V registers.
- `instr` → Stores fetched instruction.

2. Program Counter Logic

```
always @(posedge clk) begin  
    if (!resetsn)  
        pc <= 32'b0;  
    else  
        pc <= next_pc;  
end
```

Function:

- On reset, PC becomes zero.
- During execution, PC updates every clock cycle.

3. Instruction Fetch

```
always @(posedge clk) begin  
  
    mem_valid <= 1'b1;  
    mem_instr <= 1'b1;  
  
    mem_addr <= pc;  
  
end
```

Function:

- Requests instruction from memory.
- Sends address of instruction.
- **4. ALU Operation**



```
• always @(*) begin
    case (instr[6:0])
        7'b0110011:
            alu_result = reg1 + reg2;
        7'b1100011:
            alu_result = reg1 - reg2;
        default:
            alu_result = 32'b0;
    endcase
end
```

Function:

- Performs arithmetic and logical operations.
- Supports RISC-V instructions.

5. Register Write Operation

```
always @(posedge clk) begin
    if (write_enable)
        registers[rd] <= alu_result;
end
```

Function:

- Stores ALU result into destination register.
-

6. Memory Interface

```
assign mem_wdata = store_data;
```

Function:

- Sends data during store instructions.
-

7. Reset Logic

```
always @(posedge clk) begin
    if (!resetn) begin
        pc <= 0;
    end
end
```



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```
mem_valid <= 0;

end

end
```

Function:

- Initializes processor.

Program 2: config.tcl (OpenLane Constraint File)

File Name:

config.tcl

Purpose:

- config.tcl is the main configuration file used by OpenLane.
- It defines the design name, RTL source file, clock details, floorplan settings, placement settings, CTS settings, and routing options.
- OpenLane reads this file before starting the RTL-to-GDSII flow.

Location:

designs/riscv32/config.tcl

Complete config.tcl Program

```
#-----
# Design Name
#-----
```

```
set ::env(DSIGN_NAME) picorv32
```

```
#-----
# RTL Source File
#-----
```

```
set ::env(VERILOG_FILES) \
$::env(DSIGN_DIR)/src/picorv32.v
```

```
#-----
# Clock Configuration
#-----
```

```
set ::env(CLOCK_PORT) clk
```

```
set ::env(CLOCK_PERIOD) 10
```

```
#-----
# Synthesis Settings
#-----
```

```
# Clock uncertainty during synthesis
```



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15% of clock period

$10\text{ns} \times 15\% = 1.5\text{ns}$

set ::env(SYNTH_CLOCK_UNCERTAINTY) 1.5

Timing derate

3% variation

set ::env(SYNTH_TIMING_DERATE) 0.03

#-----

Input and Output Delay

#-----

50% of clock period

$10\text{ns} \times 50\% = 5\text{ns}$

set ::env(INPUT_DELAY) 5

set ::env(OUTPUT_DELAY) 5

#-----

Floorplan Settings

#-----

Core utilization = 60%

set ::env(FP_CORE_UTIL) 60

Floorplan aspect ratio = 0.7

set ::env(FP_ASPECT_RATIO) 0.7

#-----

Pin Placement Settings

#-----

Use pin order file

**set ::env(FP_PIN_ORDER_CFG) \
\$::env(DESIGN_DIR)/pin_order.cfg**

Place pins on Metal4 layer

set ::env(FP_IO_HMETAL) metal4

set ::env(FP_IO_VMETAL) metal4

#-----

Power Distribution Network

#-----



Enable core ring

```
set ::env(FP_PDN_CORE_RING) 1
```

```
#-----  
# Clock Tree Synthesis (CTS)  
#-----
```

```
# CTS target skew  
# 5% of 10ns = 0.5ns
```

```
set ::env(CTS_TARGET_SKEW) 0.5
```

Enable clock tree synthesis

```
set ::env(RUN_CTS) 1
```

```
#-----  
# Routing Settings  
#-----
```

```
set ::env(RUN_ROUTING) 1
```

Program 3: `pin_order.cfg` (Port Placement File)

`pin_order.cfg`

Purpose:

- `pin_order.cfg` is used by OpenLane during the **Floorplan stage**.
- It defines where the input and output ports should be placed on the chip boundary.
- In this assignment:
 - Inputs are placed on one side.
 - Outputs are placed on the opposite side.
 - Ports are assigned to **Metal4 layer** through `config.tcl`.

Location:

```
designs/riscv32/pin_order.cfg
```

Complete `pin_order.cfg` Program

```
#-----  
# Pin Placement Configuration  
# RISC-V 32-bit RTL  
#-----
```

```
#-----  
# North Side Pins  
#-----
```

N



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clk
resetn

#-----
South Side Pins
#-----

S

mem_valid
mem_instr
mem_addr[0]
mem_addr[1]
mem_addr[2]
mem_addr[3]
mem_addr[4]
mem_addr[5]
mem_addr[6]
mem_addr[7]
mem_addr[8]
mem_addr[9]
mem_addr[10]
mem_addr[11]
mem_addr[12]
mem_addr[13]
mem_addr[14]
mem_addr[15]
mem_addr[16]
mem_addr[17]
mem_addr[18]
mem_addr[19]
mem_addr[20]
mem_addr[21]
mem_addr[22]
mem_addr[23]
mem_addr[24]
mem_addr[25]
mem_addr[26]
mem_addr[27]
mem_addr[28]
mem_addr[29]
mem_addr[30]
mem_addr[31]

#-----
East Side Pins
#-----

E

mem_ready
mem_rdata[0]
mem_rdata[1]
mem_rdata[2]
mem_rdata[3]
mem_rdata[4]
mem_rdata[5]



```
mem_rdata[6]
mem_rdata[7]
mem_rdata[8]
mem_rdata[9]
mem_rdata[10]
mem_rdata[11]
mem_rdata[12]
mem_rdata[13]
mem_rdata[14]
mem_rdata[15]
mem_rdata[16]
mem_rdata[17]
mem_rdata[18]
mem_rdata[19]
mem_rdata[20]
mem_rdata[21]
mem_rdata[22]
mem_rdata[23]
mem_rdata[24]
mem_rdata[25]
mem_rdata[26]
mem_rdata[27]
mem_rdata[28]
mem_rdata[29]
mem_rdata[30]
mem_rdata[31]
```

```
#-----
# West Side Pins
#-----
```

```
W
mem_wdata[0]
mem_wdata[1]
mem_wdata[2]
mem_wdata[3]
mem_wdata[4]
mem_wdata[5]
mem_wdata[6]
mem_wdata[7]
mem_wdata[8]
mem_wdata[9]
mem_wdata[10]
mem_wdata[11]
mem_wdata[12]
mem_wdata[13]
mem_wdata[14]
mem_wdata[15]
mem_wdata[16]
mem_wdata[17]
mem_wdata[18]
```



```
mem_wdata[19]
mem_wdata[20]
mem_wdata[21]
mem_wdata[22]
mem_wdata[23]
mem_wdata[24]
mem_wdata[25]
mem_wdata[26]
mem_wdata[27]
mem_wdata[28]
mem_wdata[29]
mem_wdata[30]
mem_wdata[31]
```

Program 4: OpenLane TCL Flow Commands (`flow.tcl`)

Purpose:

- `flow.tcl` is the main TCL script used by OpenLane to execute the complete RTL-to-GDSII physical design flow.
- It controls all stages:
 - Synthesis
 - Floorplan
 - Placement
 - Clock Tree Synthesis (CTS)
 - Routing
 - Final GDS generation

Location:OpenLane/`flow.tcl`

. Tools Used

- Linux environment / Ubuntu
- Docker
- OpenLane
- OpenROAD
- Yosys (RTL synthesis)
- OpenSTA (Static Timing Analysis)
- Magic (Layout viewer)
- Sky130 PDK
- RISC-V 32-bit RTL source code

. Required Design Files

```
designs/
├── riscv32/
│   ├── config.tcl
│   ├── src/
│   │   └── picorv32.v
│   └── pin_order.cfg
```

. Starting OpenLane Environment

Commands:



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```
cd OpenLane
```

```
make mount
```

Enter OpenLane container:

```
./flow.tcl -interactive
```

or

```
docker run ...
```

. Preparing Design Configuration

```
designs/riscv32/config.tcl
```

Add clock information:

```
set ::env(CLOCK_PORT) clk
set ::env(CLOCK_PERIOD) 10
```

.1 Clock Uncertainty

For synthesis:

```
set_clock_uncertainty 15% of clock period
```

Example:

Clock period = 10ns

```
set_clock_uncertainty 1.5
```

For CTS:

```
set_clock_uncertainty 0.5
```

(5% of 10ns)

.2 Clock Derate

Set:

```
set_timing_derate 3%
```

Used to model variation in delay.

.3 Input and Output Delay



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50% of clock period:

For 10ns clock:

Input delay:

```
set_input_delay 5
```

Output delay:

```
set_output_delay 5
```

. Floorplan Settings

Utilization

Set:

```
FP_CORE_UTIL = 60
```

Meaning:

60% of core area is occupied by standard cells.

Floorplan Ratio

Set:

```
FP_ASPECT_RATIO = 0.7
```

. Port Placement

Requirement:

- Inputs on one side
- Outputs on opposite side
- Metal4 layer

File:

```
pin_order.cfg
```

Example:

```
N  
clk  
reset
```

```
S  
out_data
```



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Layer:

```
FP_PIN_ORDER_CFG
```

and

```
FP_IO_HLAYER = metal4  
FP_IO_VLAYER = metal4
```

. Enable Core Ring

In config.tcl:

```
FP_PDN_CORE_RING = 1
```

Purpose:

- Provides power distribution
 - Reduces IR drop
-

. CTS Configuration

Target skew:

5% of clock period

Example:

10ns clock:

```
CTS_TARGET_SKEW = 0.5
```

Enable clock network delay:

```
CTS_CLK_MAX_WIRE_LENGTH
```

. Running OpenLane Flow

Inside OpenLane:

Synthesis

```
./flow.tcl -design riscv32 -tag run1 -to synth
```

Floorplan



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```
./flow.tcl -design riscv32 -tag run1 -to floorplan
```

Placement

```
./flow.tcl -design riscv32 -tag run1 -to placement
```

CTS

```
./flow.tcl -design riscv32 -tag run1 -to cts
```

Routing

```
./flow.tcl -design riscv32 -tag run1
```

. Viewing Layout

Open Magic:

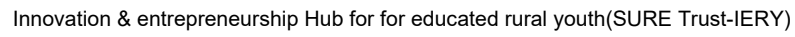
```
magic -T /pdk/sky130A/libs.tech/magic/sky130A.tech \  
designs/riscv32/runs/run1/results/final/def/
```

Check:

- Standard cells
 - Power rings
 - Clock tree
 - Routing
-

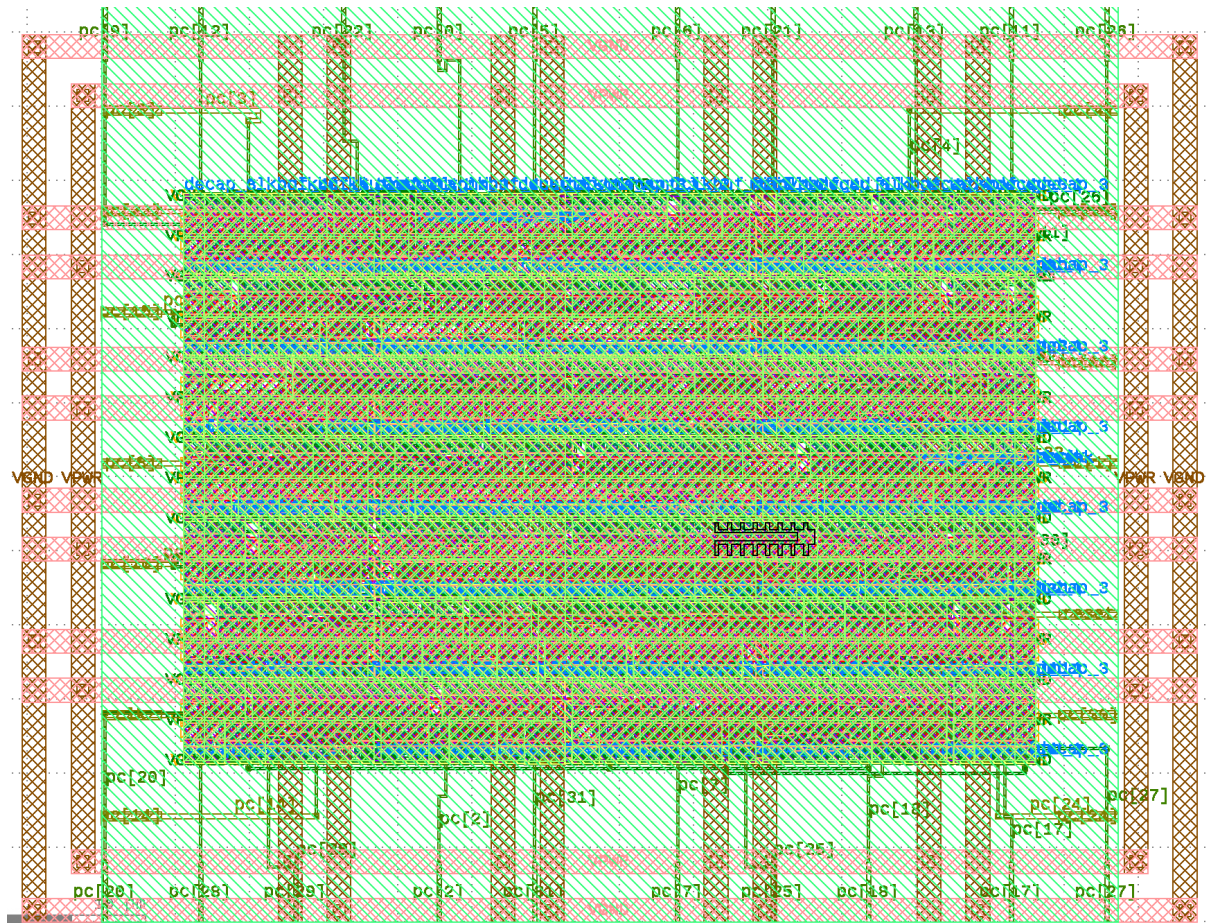
. Results :

1. Synthesis completed
 2. Floorplan view
 3. Placement view
 4. CTS clock tree
 5. Final routed layout
 6. Timing report
-





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```
Last login: Tue Jun 30 09:42:17 on console
mounika@mounika-MacBook-Air ~ % cd ~/OpenLane/designs/riscv32
mounika@mounika-MacBook-Air riscv32 % ls
config.tcl  pin_order.cfg  runs  src
mounika@mounika-MacBook-Air riscv32 % nano config.tcl
mounika@mounika-MacBook-Air riscv32 % mkdir -p scripts
nano scripts/post_cts.tcl
mounika@mounika-MacBook-Air riscv32 % cat scripts/post_cts.tcl
set_propagated_clock [all_clocks]
mounika@mounika-MacBook-Air riscv32 % nano config.tcl
mounika@mounika-MacBook-Air riscv32 % cd ~/OpenLane
rm -rf designs/riscv32/runs
mounika@mounika-MacBook-Air OpenLane % make mount
cd /Users/mounika/OpenLane && \
  docker run --rm -v /Users/mounika/OpenLane:openlane -v /Users/mounika/OpenLane/designs:openlane/install -v /Users/mounika:/Users/mounika -v /Users/mounika/.ciel:/Users/mounika/.ciel -e POK_R
OOT=/Users/mounika/.ciel -e POK=sky130A --user 501:20 -ti ghcr.io/the-openroad-project/openlane:ff5689f66b17bfa4068d5336495ab1718987ff69-arm64v8
OpenLane Container (ff5689f):/openlane$ ./flow.tcl -design riscv32
OpenLane v1.0.2 (ff5689f66b17bfa4068d5336495ab1718987ff69)
All rights reserved. (c) 2020-2025 Efabless Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.
```



```
mounika@mounikas-MacBook-Air OpenLane % cd ~/OpenLane/designs/riscv32
nano config.tcl
mounika@mounikas-MacBook-Air riscv32 % cat config.tcl
set ::env(DSIGN_NAME) riscv32

set ::env(VERILOG_FILES) \
$::env(DSIGN_DIR)/src/riscv32.v
[

# Clock
set ::env(CLOCK_PORT) clk
set ::env(CLOCK_PERIOD) 10

# Clock uncertainty
set ::env(SYNTH_CLK_UNCERTAINTY) 1.5
set ::env(CTS_CLK_UNCERTAINTY) 0.5

# Timing derate
set ::env(SYNTH_TIMING_DERATE) 3

# Input and output delay
set ::env(INPUT_DELAY) 5
set ::env(OUTPUT_DELAY) 5

# Floorplan
set ::env(FP_CORE_UTIL) 60
set ::env(FP_ASPECT_RATIO) 0.7

# IO metal layer
set ::env(FP_IO_HMETAL_LAYER) met4
set ::env(FP_IO_VMETAL_LAYER) met4

# Core ring
set ::env(FP_PDN_CORE_RING) 1

# CTS
set ::env(CTS_TARGET_SKEW) 0.5

# Post CTS
set ::env(POST_CTS_TCL) $::env(DSIGN_DIR)/scripts/post_cts.tcl
```



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```
mounika@mounikas-MacBook-Air OpenLane % cd ~/OpenLane
nano designs/riscv32/config.tcl
mounika@mounikas-MacBook-Air OpenLane % cat designs/riscv32/config.tcl
set ::env(DSIGN_NAME) riscv32

set ::env(VERILOG_FILES) \
$::env(DSIGN_DIR)/src/riscv32.v

# Clock
set ::env(CLOCK_PORT) clk
set ::env(CLOCK_PERIOD) 10

# Clock uncertainty
set ::env(SYNTH_CLK_UNCERTAINTY) 1.5
set ::env(CTS_CLK_UNCERTAINTY) 0.5

# Timing derate
set ::env(SYNTH_TIMING_DERATE) 3

# Input and output delay
set ::env(INPUT_DELAY) 5
set ::env(OUTPUT_DELAY) 5

# Floorplan
set ::env(FP_CORE_UTIL) 60
set ::env(FP_ASPECT_RATIO) 0.7

# IO metal layer
set ::env(FP_IO_HMETAL_LAYER) met4
set ::env(FP_IO_VMETAL_LAYER) met4

# Core ring
set ::env(FP_PDN_CORE_RING) 1

# CTS
set ::env(CTS_TARGET_SKEW) 0.5

# Post CTS
set ::env(POST_CTS_TCL) $::env(DSIGN_DIR)/scripts/post_cts.tcl

# Hold fixing
set ::env(PL_RESIZER_HOLD_SLACK_MARGIN) 0.1
set ::env(PL_RESIZER_HOLD_MAX_BUFFER_PERCENT) 50
set ::env(GLB_RESIZER_HOLD_MAX_BUFFER_PERCENT) 50
# Hold fixing settings
set ::env(PL_RESIZER_HOLD_SLACK_MARGIN) 0.2
set ::env(PL_RESIZER_HOLD_MAX_BUFFER_PERCENT) 80
set ::env(GLB_RESIZER_HOLD_MAX_BUFFER_PERCENT) 80

# Enable timing repair
set ::env(PL_RESIZER_TIMING_OPTIMIZATIONS) 1
set ::env(GLB_RESIZER_TIMING_OPTIMIZATIONS) 1
# Disable strict timing failure for hold
set ::env(QUIT_ON_HOLD_VIOLATIONS) 0
mounika@mounikas-MacBook-Air OpenLane %
```



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```
OpenLane Container (ff5589f):/openlane$ cd designs/riscv32/runs/
OpenLane Container (ff5589f):/openlane$ ./flow.tcl -design riscv32
OpenLane v1.0.2 (ff5589f65b17bfa406d5336495ab1718987ff69)
All rights reserved. (c) 2020-2025 Effable Corporation and contributors.
Available under the Apache License, version 2.0. See the LICENSE file for more details.

[INFO]: Using configuration in 'designs/riscv32/config.tcl'...
[INFO]: PDK Root: /Users/mounika/.ciel
[INFO]: Process Design Kit: sky130A
[INFO]: Standard Cell Library: sky130_fd_sc_hd
[INFO]: Optimization Standard Cell Library: sky130_fd_sc_hd
[INFO]: Run Directory: /openlane/designs/riscv32/runs/RUN_2026.07.01_16.52.12
[INFO]: Saving runtime environment...
[INFO]: Preparing LEF files for the nom corner...
[INFO]: Preparing LEF files for the min corner...
[INFO]: Preparing LEF files for the max corner...
[INFO]: Running linter (Verilator) (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/synthesis/linter.log)...
[INFO]: 0 errors found by linter
[INFO]: 0 warnings found by linter
[STEP 1]
[INFO]: Running Synthesis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/synthesis/1-synthesis.log)...
[STEP 2]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/synthesis/2-sta.log)...
[STEP 3]
[INFO]: Running Initial Floorplanning (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/floorplan/3-initial_fp.log)...
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
[INFO]: Floorplanned with width 52.44 and height 51.68.
[STEP 4]
[INFO]: Running IO Placement...
[STEP 5]
[INFO]: Running Tap/Decap Insertion (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/floorplan/5-tap.log)...
[INFO]: Power planning with power {VPWR} and ground {VGND}...
[STEP 6]
[INFO]: Generating PDN (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/floorplan/6-pdn.log)...
[STEP 7]
[INFO]: Running Global Placement (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/placement/7-global.log)...
[STEP 8]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/placement/8-gpi_sta.log)...
[STEP 9]
[INFO]: Running Placement Resizer Design Optimizations (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/placement/9-resizer.log)...
[STEP 10]
[INFO]: Running Detailed Placement (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/placement/10-detailed.log)...
[STEP 11]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/placement/11-dpl_sta.log)...
[STEP 12]
[INFO]: Running Clock Tree Synthesis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/cts/12-cts.log)...
[STEP 13]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/cts/13-cts_sta.log)...
[STEP 14]
[INFO]: Running Placement Resizer Timing Optimizations (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/cts/14-resizer.log)...
[STEP 15]
[INFO]: Running Global Routing Resizer Design Optimizations (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/15-resizer_design.log)...
[STEP 16]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/16-rsz_design_sta.log)...
[STEP 17]
[INFO]: Running Global Routing Resizer Timing Optimizations (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/17-resizer_timing.log)...
[STEP 18]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/18-rsz_timing_sta.log)...
[STEP 19]
[INFO]: Running Global Routing (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/19-global.log)...
[INFO]: Starting OpenROAD Antenna Repair Iterations...
[STEP 20]
[INFO]: Writing Verilog (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/19-global_write_netlist.log)...
[STEP 21]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/21-grt_sta.log)...
[STEP 21]
[INFO]: Running Single-Corner Static Timing Analysis (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/21-grt_sta.log)...
[STEP 22]
[INFO]: Running Fill Insertion (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/22-fill.log)...
[STEP 23]
[INFO]: Running Detailed Routing (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/23-detailed.log)...
[INFO]: No DRC violations after detailed routing.
[STEP 24]
[INFO]: Checking Wire Lengths (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/routing/24-wire_lengths.log)...
[STEP 25]
[INFO]: Running SPFEF Extraction at the min process corner (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/25-parasitics_extraction.min.log)...
[STEP 26]
[INFO]: Running Multi-Corner Static Timing Analysis at the min process corner (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/26-rcx_msta.min.log)...
[STEP 27]
[INFO]: Running SPFEF Extraction at the max process corner (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/27-parasitics_extraction.max.log)...
[STEP 28]
[INFO]: Running Multi-Corner Static Timing Analysis at the max process corner (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/28-rcx_msta.max.log)...
[STEP 29]
[INFO]: Running SPFEF Extraction at the nom process corner (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/29-parasitics_extraction.nom.log)...
[STEP 30]
[INFO]: Running Multi-Corner Static Timing Analysis at the nom process corner (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/30-rcx_msta.nom.log)...
[STEP 31]
[INFO]: Running Single-Corner Static Timing Analysis at the nom process corner (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/31-rcx_sta.log)...
[WARNING]: Module sky130_fd_sc_hd__tapwrvrvgnd_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_2 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_1 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd__decap_32 blackboxed during sta
[STEP 32]
[INFO]: Creating IR Drop Report (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/32-irdrop.log)...
[WARNING]: VSRC_LOC_FILES is not defined. The IR drop analysis will run, but the values may be inaccurate.
[STEP 33]
[INFO]: Running Magic to generate various views...
[INFO]: Streaming out GDSII with Magic (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/33-gdsii.log)...
[INFO]: Generating MAGICF views...
[INFO]: Generating lef with Magic (/openlane/designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/33-lef.log)...
[STEP 34]
[INFO]: Streaming out GDSII with Klayout (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/34-gdsii-klayout.log)...
[STEP 35]
[INFO]: Running XOR on the layouts using Klayout (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/35-xor.log)...
[INFO]: No XOR differences between Klayout and Magic gds.
[STEP 36]
[INFO]: Running Magic Spice Export from LEF (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/36-spice.log)...
[STEP 37]
[INFO]: Writing Powered Verilog (logs: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/37-write_powered_def.log, designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/37-write_powered_verilog.log)...
[STEP 38]
[INFO]: Writing Verilog (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/37-write_powered_verilog.log)...
[STEP 39]
[INFO]: Running LVS (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/39-lvs-lef.log)...
[STEP 40]
[INFO]: Running Magic DRC (log: designs/riscv32/runs/RUN_2026.07.01_16.52.12/logs/signoff/40-drc.log)...
[INFO]: Converting Magic DRC database to various tool-readable formats...
[INFO]: No DRC violations after DRC streaming out.
```



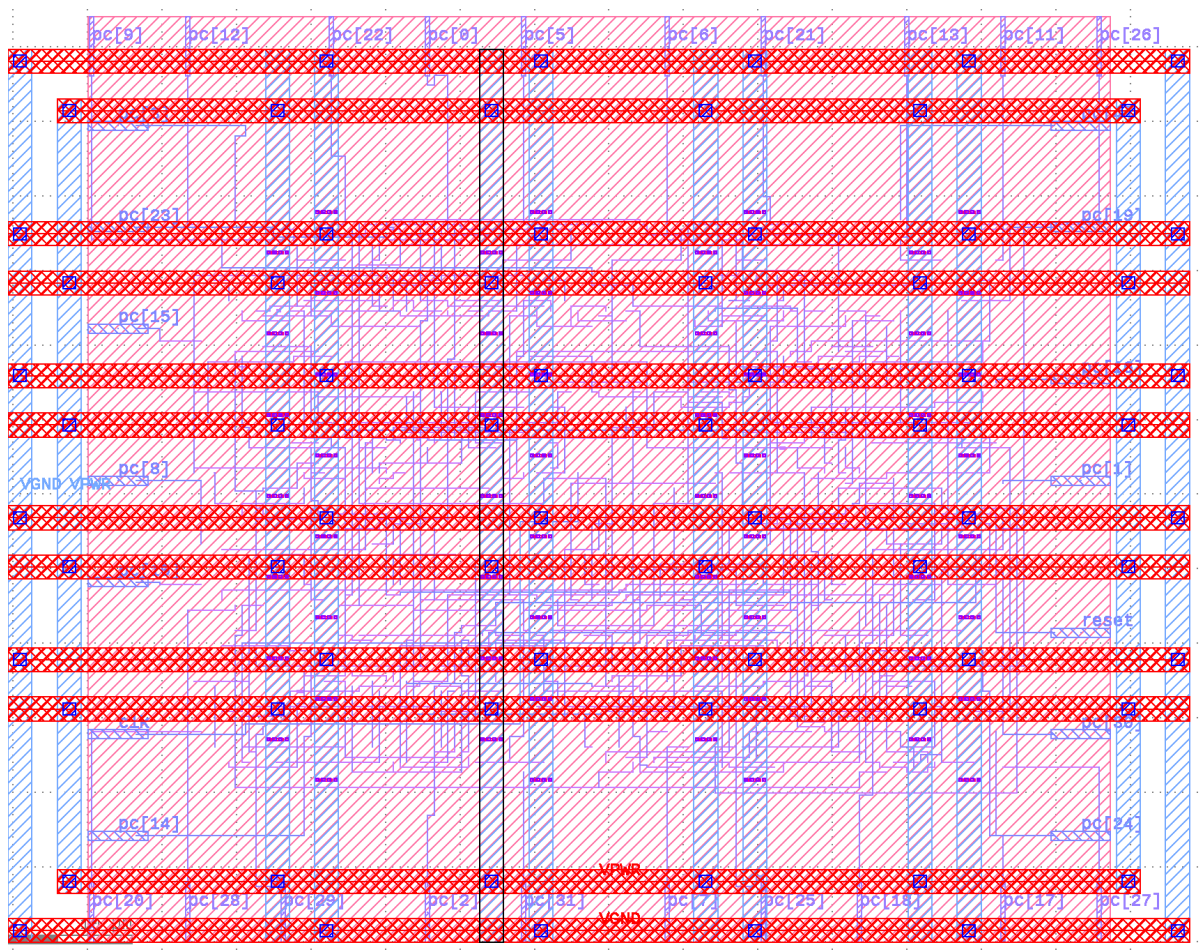
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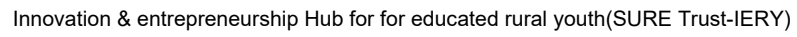
```
[STEP 26]
[INFO]: Running Multi-Corner Static Timing Analysis at the min process corner (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/26-rcx_msta.min.log)...
[STEP 27]
[INFO]: Running SPEF Extraction at the max process corner (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/27-parasitics_extraction.max.log)...
[STEP 28]
[INFO]: Running Multi-Corner Static Timing Analysis at the max process corner (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/28-rcx_msta.max.log)...
[STEP 29]
[INFO]: Running SPEF Extraction at the nom process corner (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/29-parasitics_extraction.nom.log)...
[STEP 30]
[INFO]: Running Multi-Corner Static Timing Analysis at the nom process corner (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/30-rcx_msta.nom.log)...
[STEP 31]
[INFO]: Running Single-Corner Static Timing Analysis at the nom process corner (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/31-rcx_sta.log)...
[WARNING]: Module sky130_fd_sc_hd__tapvprvrgnd_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_2 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_1 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd__decap_12 blackboxed during sta
[STEP 32]
[INFO]: Creating IR Drop Report (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/32-irdrop.log)...
[WARNING]: VSRC_LOC_FILES is not defined. The IR drop analysis will run, but the values may be inaccurate.
[STEP 33]
[INFO]: Running Magic to generate various views...
[INFO]: Streaming out GDSII with Magic (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/33-gdsii.log)...
[INFO]: Generating MAGLEF views...
[INFO]: Generating lef with Magic (/openlane/designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/33-lef.log)...
[STEP 34]
[INFO]: Streaming out GDSII with KLayout (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/34-gdsii-klayout.log)...
[STEP 35]
[INFO]: Running XOR on the layouts using KLayout (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/35-xor.log)...
[INFO]: No XOR differences between Klayout and Magic gds.
[STEP 36]
[INFO]: Running Magic Spice Export from LEF (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/36-spice.log)...
[STEP 37]
[INFO]: Writing Powered Verilog (logs: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/37-write_powered_def.log, designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/37-write_powered_verilog.log)...
[STEP 38]
[INFO]: Writing Verilog (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/37-write_powered_verilog.log)...
[STEP 39]
[INFO]: Running LVS (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/39-lvs.lef.log)...
[STEP 40]
[INFO]: Running Magic DRC (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/40-drc.log)...
[INFO]: Converting Magic DRC database to various tool-readable formats...
[INFO]: No DRC violations after GDS streaming out.
[STEP 41]
[INFO]: Running OpenROAD Antenna Rule Checker (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/41-arc.log)...
[STEP 42]
[INFO]: Running Circuit Validity Checker ERC (log: designs/riscv32/runs/RUN_2026.07.01.16.52.12/logs/signoff/42-erc_screen.log)...
[INFO]: Saving current set of views in 'designs/riscv32/runs/RUN_2026.07.01.16.52.12/results/final'...
[INFO]: Saving runtime environment...
[INFO]: Generating final set of reports...
[INFO]: Created manufacturability report at 'designs/riscv32/runs/RUN_2026.07.01.16.52.12/reports/manufacturability.rpt'.
[INFO]: Created metrics report at 'designs/riscv32/runs/RUN_2026.07.01.16.52.12/reports/metrics.csv'.
[INFO]: There are no max slew, max fanout or max capacitance violations in the design at the typical corner.
[WARNING]: There are hold violations in the design at the typical corner. Please refer to 'designs/riscv32/runs/RUN_2026.07.01.16.52.12/reports/signoff/31-rcx_sta.min.rpt'.
[INFO]: There are no setup violations in the design at the typical corner.
SUCCESS: Flow complete.
[INFO]: Note that the following warnings have been generated:
[WARNING]: Current core area is too small for the power grid settings chosen. The power grid will be scaled down.
[WARNING]: Module sky130_fd_sc_hd__tapvprvrgnd_1 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_2 blackboxed during sta
[WARNING]: Module sky130_fd_sc_hd__fill_1 blackboxed during sta
[WARNING]: Module sky130_ef_sc_hd__decap_12 blackboxed during sta
[WARNING]: VSRC_LOC_FILES is not defined. The IR drop analysis will run, but the values may be inaccurate.
[WARNING]: There are hold violations in the design at the typical corner. Please refer to 'designs/riscv32/runs/RUN_2026.07.01.16.52.12/reports/signoff/31-rcx_sta.min.rpt'.

OpenLane Container (ff5609f):/openlane$ ./flow.tcl -design riscv32
```



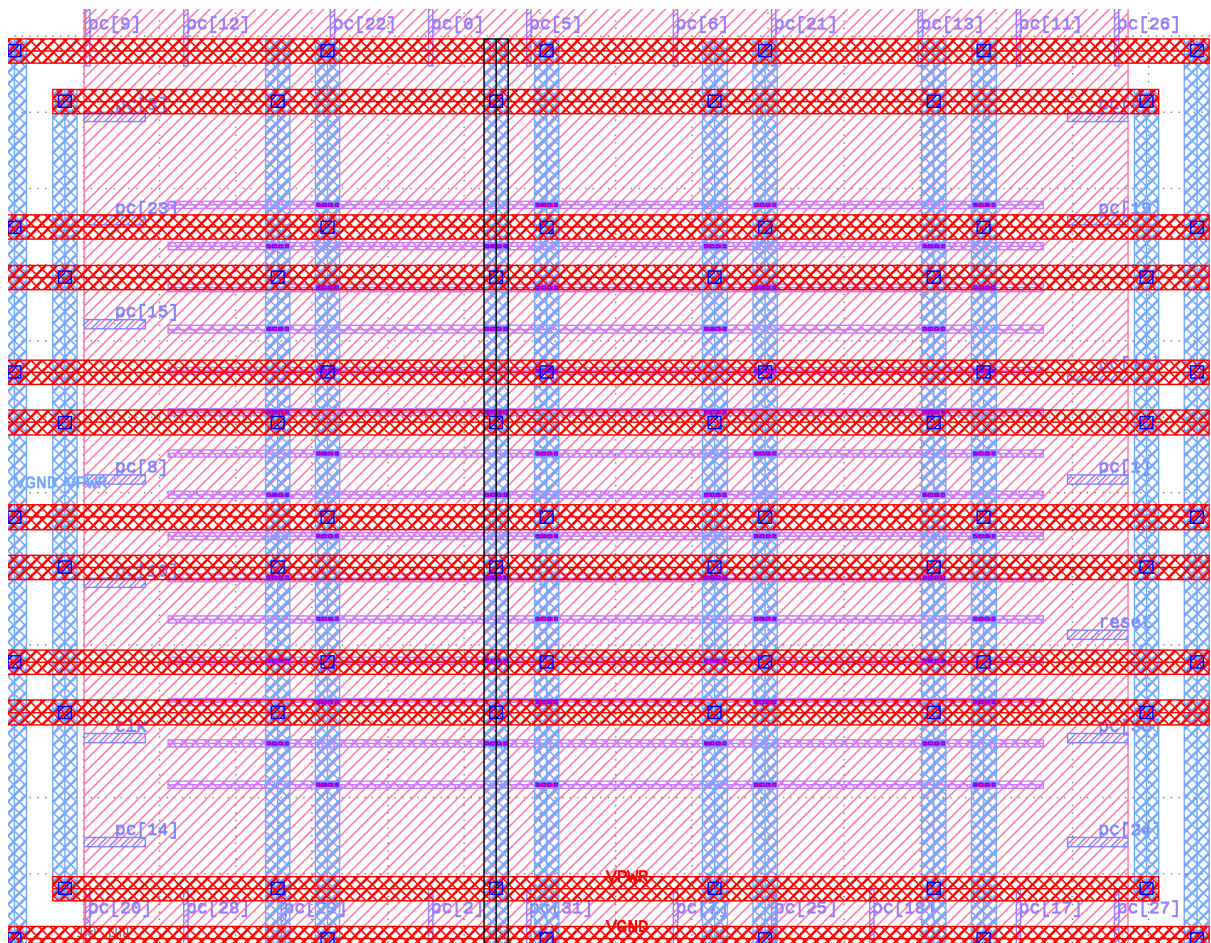
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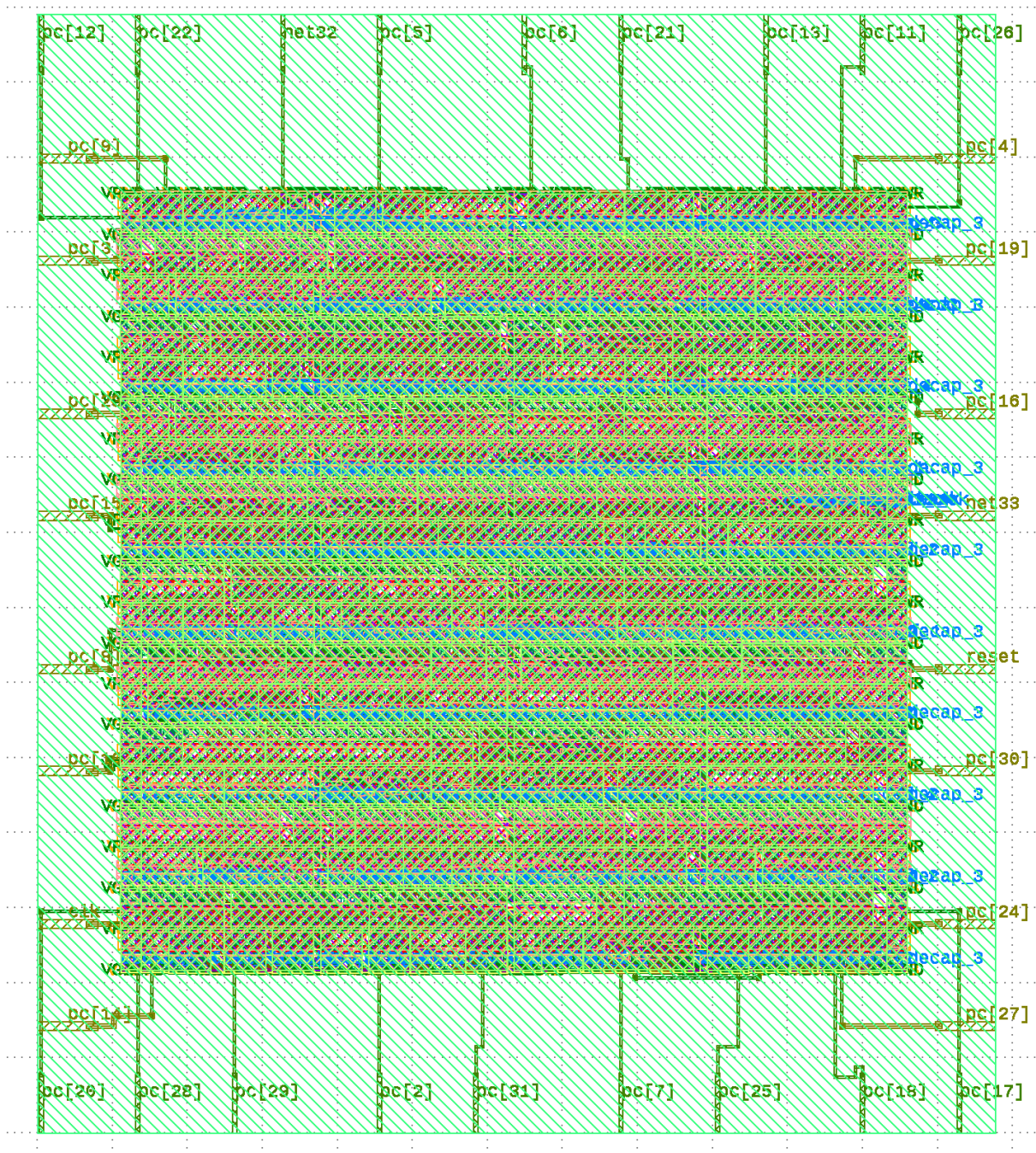


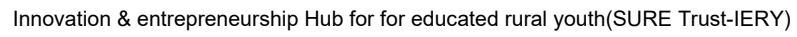


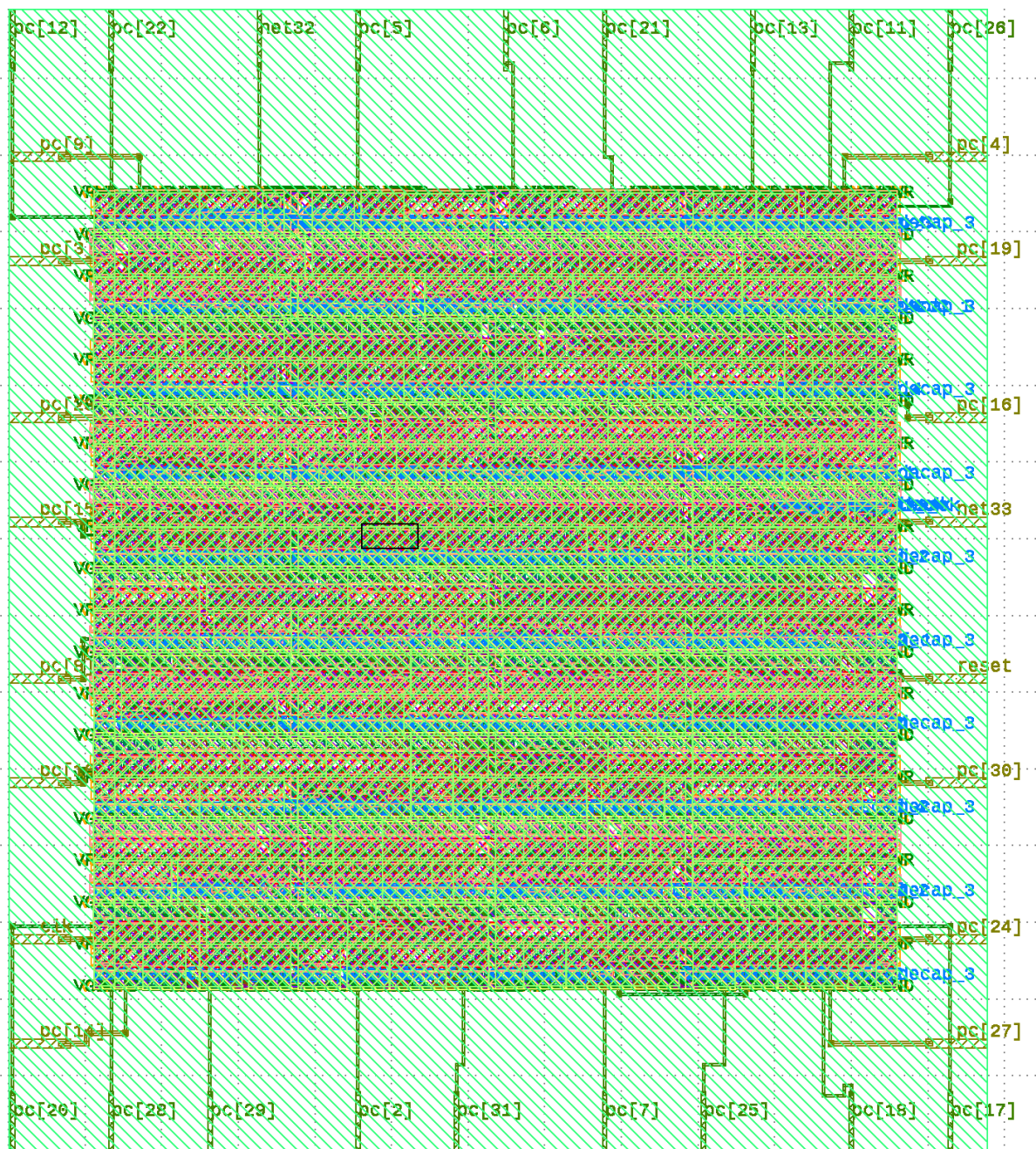
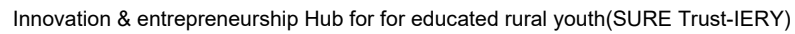


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OpenLane Setting

- ```
SYNTH_CLOCK_UNCERTAINTY
CTS_CLK_UNCERTAINTY
PL_SET_PE_CORE / OpenSTA derate setting
SDC constraints
FP CORE UTIL = 60
```



### Requirement

5. Floorplan ratio = 0.7

6. Inputs one side, outputs other side, Metal4

7. Enable core ring

8. CTS target skew = 5% clock period

9. Clock network delay inserted after CTS

### OpenLane Setting

FP\_ASPECT\_RATIO = 0.7

FP\_PIN\_ORDER\_CFG + FP\_IO\_HLAYER = met4

FP\_PDN\_CORE\_RING = 1

CTS\_TARGET\_SKEW

CTS\_CLK\_MAX\_WIRE\_LENGTH / post CTS settings

---

results/

|             |                             |
|-------------|-----------------------------|
| — synthesis | → RTL to gate-level netlist |
| — floorplan | → Chip/core planning        |
| — placement | → Standard cell placement   |
| — cts       | → Clock Tree Synthesis      |
| — routing   | → Metal routing             |
| — signoff   | → Final checks              |
| — final     | → Final ASIC outputs        |

|         |                                                   |
|---------|---------------------------------------------------|
| def     | → Final physical layout description               |
| gds     | → Final chip layout database (fabrication format) |
| lef     | → Library exchange format                         |
| mag     | → Magic layout files                              |
| sdc     | → Timing constraints                              |
| spef    | → Parasitic extraction data                       |
| verilog | → Final netlist                                   |
| lib     | → Standard cell libraries                         |
| sdf     | → Timing delay data                               |
| spi     | → Spice files                                     |

## 2>OpenLane Physical Design Analysis of RISC-V Using KLayout

### Aim

To analyze the different stages of ASIC physical design flow (Floorplan, Placement, CTS, and Routing) using OpenLane generated files and visualize the final layout using KLayout.

### Tools Used

- OpenLane
  - SKY130 PDK
  - KLayout
  - Docker Environment
-



# 1. Floorplan Analysis

## Objective

To analyze chip dimensions, core area, and physical-only cells.

### Input File:

```
results/floorplan/riscv32.def
```

## Measurements

| Parameter           | Value                |
|---------------------|----------------------|
| Die Width           | 68.62 $\mu\text{m}$  |
| Die Height          | 62.065 $\mu\text{m}$ |
| Core Width          | 57.5 $\mu\text{m}$   |
| Core Height         | 62.56 $\mu\text{m}$  |
| Physical-only Cells | 60                   |

## Physical-only Cells Identified

- Tap cells
- Decap cells
- Filler cells
- Endcap cells

These cells improve power integrity, reliability, and manufacturing requirements.

---

# 2. Placement Analysis

## Objective

To analyze standard cell placement and identify buffers, inverters, and flip-flops.

### Input File:

```
results/placement/riscv32.def
```

## Cell Analysis



### Cell Type Count

Buffers 22

Inverters 0

Flip-Flops 30

### Site Row Measurement

| Parameter       | Value              |
|-----------------|--------------------|
| Site Row Width  | 57.5 $\mu\text{m}$ |
| Site Row Height | 2.72 $\mu\text{m}$ |

Placement arranges standard cells in rows while optimizing timing and area.

---

## 3. Clock Tree Synthesis (CTS) Analysis

### Objective

To analyze clock tree generation and inserted clock buffers.

### Input File:

results/cts/riscv32.def

### CTS Cell Analysis

| Cell Type                   | Count |
|-----------------------------|-------|
| Clock Buffers               | 30    |
| Total CTS Buffers/Inverters | 46    |

CTS inserts clock buffers to:

- Reduce clock skew
- Improve clock transition
- Balance clock distribution

---

## 4. Routing Analysis

### Objective

To analyze final signal routing and optimization cells.



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### Input File:

results/routing/riscv32.def

### Routing Analysis

| Parameter                 | Value |
|---------------------------|-------|
| Routing Buffers/Inverters | 11    |
| Vias                      | 6     |

Routing connects all placed cells using:

- Metal layers
- Vias
- Interconnect paths

---

## Final Layout Visualization

The final GDSII file:

results/signoff/riscv32.gds

was opened using **KLayout**.

KLayout was used to:

- View complete chip layout
- Zoom into routing details
- Observe different physical layers

---

## Result

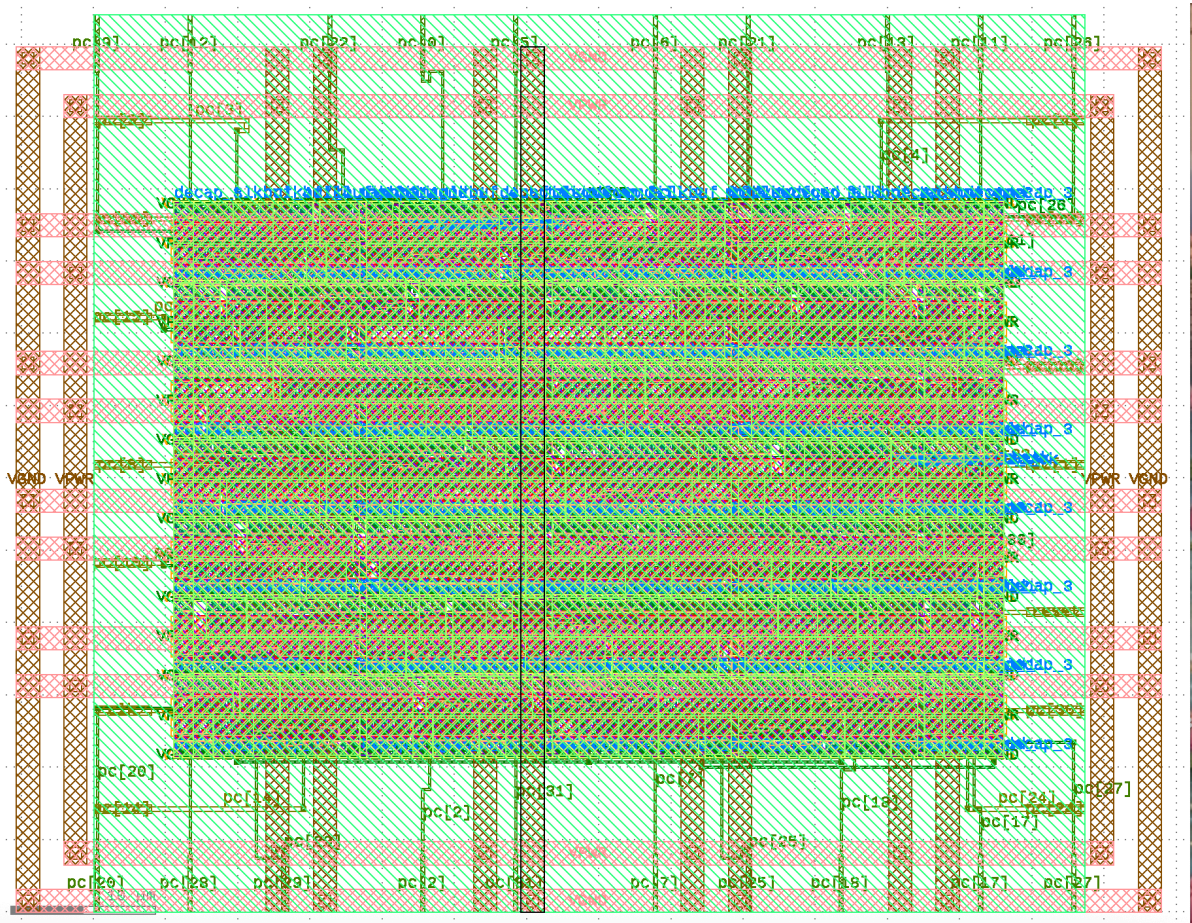
The RISC-V design was successfully analyzed through all backend physical design stages. Floorplan dimensions, placement cells, CTS clock buffers, routing elements, and final GDS layout were verified.

## Conclusion

OpenLane successfully generated the physical design of the RISC-V processor. KLayout provided a simple method to inspect the final ASIC layout and understand different stages of the VLSI design flow.



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## Innovation & entrepreneurship Hub for for educated rural youth(SURE Trust-IERy)

```
mounika@mounikas-MacBook-Air OpenLane % grep -E "tap|decap|fill|filler|endcap" designs/riscv32/runs/run1/results/floorplan/riscv32.def | wc -l
60
mounika@mounikas-MacBook-Air OpenLane % grep -r "sky130_fd_sc_hd_and" /pdk/sky130A/libs.ref/sky130_fd_sc_hd/
grep: /pdk/sky130A/libs.ref/sky130_fd_sc_hd/: No such file or directory
mounika@mounikas-MacBook-Air OpenLane % head -20 designs/riscv32/runs/run1/results/floorplan/riscv32.def
VERSION 5.8 ;
DIVIDERCHAR "/" ;
BUSBITCHARS "{}" ;
DESIGN riscv32 ;
UNITS DISTANCE MICRONS 1000 ;
DIEAREA (0 0) (68620 62065) ;
ROW ROW_0 unithd 5520 10880 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_1 unithd 5520 13600 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_2 unithd 5520 16320 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_3 unithd 5520 19040 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_4 unithd 5520 21760 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_5 unithd 5520 24480 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_6 unithd 5520 27200 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_7 unithd 5520 29920 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_8 unithd 5520 32640 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_9 unithd 5520 35360 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_10 unithd 5520 38080 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_11 unithd 5520 40800 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_12 unithd 5520 43520 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_13 unithd 5520 46240 FS DO 125 BY 1 STEP 460 0 ;
mounika@mounikas-MacBook-Air OpenLane % grep -i "DIEAREA" designs/riscv32/runs/run1/results/floorplan/riscv32.def
DIEAREA (0 0) (68620 62065) ;
mounika@mounikas-MacBook-Air OpenLane % grep -E "sky130_fd_sc_hd__(buf|inv)" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
22
mounika@mounikas-MacBook-Air OpenLane % grep "sky130_fd_sc_hd__buf" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
22
mounika@mounikas-MacBook-Air OpenLane % grep "sky130_fd_sc_hd__inv" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
0
mounika@mounikas-MacBook-Air OpenLane % grep "sky130_fd_sc_hd__df" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
30
mounika@mounikas-MacBook-Air OpenLane % grep -E "sky130_fd_sc_hd__(buf|inv)" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
22
mounika@mounikas-MacBook-Air OpenLane % grep "sky130_fd_sc_hd__df" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
30
mounika@mounikas-MacBook-Air OpenLane % find ~/OpenLane -name "sky130_fd_sc_hd__buf_1.lef"
mounika@mounikas-MacBook-Air OpenLane %
mounika@mounikas-MacBook-Air OpenLane % head -20 designs/riscv32/runs/run1/results/floorplan/riscv32.def
VERSION 5.8 ;
DIVIDERCHAR "/" ;
BUSBITCHARS "{}" ;
DESIGN riscv32 ;
UNITS DISTANCE MICRONS 1000 ;
DIEAREA (0 0) (68620 62065) ;
ROW ROW_0 unithd 5520 10880 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_1 unithd 5520 13600 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_2 unithd 5520 16320 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_3 unithd 5520 19040 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_4 unithd 5520 21760 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_5 unithd 5520 24480 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_6 unithd 5520 27200 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_7 unithd 5520 29920 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_8 unithd 5520 32640 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_9 unithd 5520 35360 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_10 unithd 5520 38080 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_11 unithd 5520 40800 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_12 unithd 5520 43520 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_13 unithd 5520 46240 FS DO 125 BY 1 STEP 460 0 ;
mounika@mounikas-MacBook-Air OpenLane % grep -i "CORE" designs/riscv32/runs/run1/results/floorplan/riscv32.def
mounika@mounikas-MacBook-Air OpenLane % grep -i "ROW" designs/riscv32/runs/run1/results/floorplan/riscv32.def | head
ROW ROW_0 unithd 5520 10880 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_1 unithd 5520 13600 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_2 unithd 5520 16320 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_3 unithd 5520 19040 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_4 unithd 5520 21760 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_5 unithd 5520 24480 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_6 unithd 5520 27200 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_7 unithd 5520 29920 FS DO 125 BY 1 STEP 460 0 ;
ROW ROW_8 unithd 5520 32640 N DO 125 BY 1 STEP 460 0 ;
ROW ROW_9 unithd 5520 35360 FS DO 125 BY 1 STEP 460 0 ;
mounika@mounikas-MacBook-Air OpenLane % grep -E "tap|decap|fill|filler|endcap" designs/riscv32/runs/run1/results/floorplan/riscv32.def | wc -l
60
mounika@mounikas-MacBook-Air OpenLane % find ~/OpenLane -name "sky130_fd_sc_hd__buf_1.lef"
mounika@mounikas-MacBook-Air OpenLane % grep -E "sky130_fd_sc_hd__(buf|inv)" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
22
mounika@mounikas-MacBook-Air OpenLane % grep "sky130_fd_sc_hd__buf" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
22
mounika@mounikas-MacBook-Air OpenLane % grep "sky130_fd_sc_hd__inv" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
0
mounika@mounikas-MacBook-Air OpenLane % grep "sky130_fd_sc_hd__df" designs/riscv32/runs/run1/results/placement/riscv32.def | wc -l
30
mounika@mounikas-MacBook-Air OpenLane % grep "clkbuf" designs/riscv32/runs/run1/results/cts/riscv32.def | wc -l
30
mounika@mounikas-MacBook-Air OpenLane % grep -E "sky130_fd_sc_hd__(clkbuf|buf|inv)" designs/riscv32/runs/run1/results/cts/riscv32.def | wc -l
46
mounika@mounikas-MacBook-Air OpenLane % designs/riscv32/runs/run1/results/cts/riscv32.def
zsh: permission denied: designs/riscv32/runs/run1/results/cts/riscv32.def
mounika@mounikas-MacBook-Air OpenLane % grep "clkbuf" designs/riscv32/runs/run1/results/cts/riscv32.def | wc -l
30
mounika@mounikas-MacBook-Air OpenLane % grep -E "sky130_fd_sc_hd__(clkbuf|buf|inv)" designs/riscv32/runs/run1/results/cts/riscv32.def | wc -l
46
mounika@mounikas-MacBook-Air OpenLane % grep -E "sky130_fd_sc_hd__(buf|inv)" designs/riscv32/runs/run1/results/routing/riscv32.def | wc -l
11
mounika@mounikas-MacBook-Air OpenLane % grep "VIA" designs/riscv32/runs/run1/results/routing/riscv32.def | wc -l
6
mounika@mounikas-MacBook-Air OpenLane % grep "^ - " designs/riscv32/runs/run1/results/routing/riscv32.def | wc -l
0
mounika@mounikas-MacBook-Air OpenLane %
```